DAOFU ZHANG

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EDUCATION

Ph.D., Electrical and Computer Engineering 2024 – Present
University of Utah · *Advisor: Prof. Fuheng Zhao*

M.S., Data Science and Machine Learning 03/2023
UC San Diego

B.E., Computer Science and Engineering 07/2021
The Hong Kong University of Science and Technology (HKUST) · *First Class Honors*

RESEARCH INTERESTS

- **LLM Model Routing and Cost-Efficient Inference:** budget-constrained model selection for LLM-powered data processing and agent workloads; task-aligned routing from small paired samples; batch and streaming routing with spending guarantees.
- **Conformal Inference and Novelty Detection:** finite-sample FDR control for ML-based detectors built on conformal p-values; adversarial robustness of these guarantees.
- **Multiple Testing in Distributed Systems:** FDR control in distributed hypothesis testing; Byzantine attacks on reported p-values and counter-attacks that restore FDR control.

PUBLICATIONS AND MANUSCRIPTS

Conference Papers

[C1] **D. Zhang**, M. Pournaderi, Y. Xiang, and P. K. Varshney, "Distributed Multiple Testing with False Discovery Rate Control in the Presence of Byzantines," in *Proc. IEEE International Symposium on Information Theory (ISIT)*, Ann Arbor, MI, USA, June 2025, pp. 1–6. doi:10.1109/ISIT63088.2025.11195318

Under Review

[S2] **D. Zhang**, Q. Zeng, C. Liu, and F. Zhao, "ARoute: Task-Aligned Batch and Streaming Model Routing," submitted to *Proceedings of the VLDB Endowment (PVLDB)*, 2027. **Submitted.**

[S1] **D. Zhang**, M. Pournaderi, H. M. Clifford, Y. Xiang, and P. K. Varshney, "On the Adversarial Robustness of Learning-based Conformal Novelty Detection," *Communications in Statistics – Theory and Methods*. **Under review.** arXiv:2510.00463

RESEARCH EXPERIENCE

Ph.D. Research Assistant — University of Utah 2024 – Present

Task-Aligned Model Routing for LLM Workloads (ARoute) [S2] 2026

- Proposed ARoute, a router that learns per-request accuracy-gain and incremental-cost estimates to choose between a weaker and a stronger LLM *before* inference, adapting to a new task and model pair from about 50 paired records.
- Designed two variants: ARoute-Batch assigns the stronger model to the requests with the greatest expected benefit under a cost budget; ARoute-Stream routes online with a validation-calibrated threshold that adjusts as the budget is consumed. Proved a finite-sample guarantee on expected normalized spending for ARoute-Stream.
- Evaluated on MMLU-Pro, LiveCodeBench, SuperGPQA, and two SWAN value-filling tasks against RouteLLM, FrugalGPT, BARGAIN, and OctoSelector: highest mean accuracy in all 15 task–cost settings (+4.1 points on average, up to +11.9) and up to 2.8× faster batch completion than cascades.

Adversarial Robustness of Conformal Novelty Detection [S1] 2025 – 2026

Supported in part by NSF Grant CCF-2611415

- Studied two learning-based conformal novelty detection frameworks with finite-sample FDR control (AdaDetect and a one-class classifier approach), analyzing how their ML-based empirical p-values behave under adversarial perturbation.
- Formulated an oracle attack and derived an upper bound on the worst-case FDR degradation; turned it into a practical attack that needs only query access to output labels.
- Combined the attack with two black-box adversarial algorithms on synthetic and real-world data, showing that FDR can be pushed well above its target level while detection power stays high.

Distributed Multiple Testing under Byzantine Attacks [C1] 2024 – 2025

Supported in part by NSF Grant CCF-2420146

- Theorized and empirically evaluated distributed multiple testing with FDR control when captured nodes report corrupted p-values.
- Characterized the effect on global FDR of two attacks: one with full knowledge of the true nulls, and one that infers nulls with the Benjamini–Hochberg procedure.
- Designed counter-attack strategies that restore FDR guarantees.

Undergraduate Researcher 04/2020 – 12/2020

A Semi-Supervised Approach to 3D Human Pose Estimation

- Generated synthetic datasets and pre-trained an auto-encoder with skip connections (PyTorch).
- Integrated few-shot layers; applied non-maximum suppression for precise joint heatmaps.

TECHNICAL SKILLS

Languages	Python, MATLAB, SQL, Java, JavaScript/TypeScript, C#, LaTeX
Deep Learning & NLP	PyTorch; Hugging Face Transformers, Datasets, Accelerate, Evaluate; fine-tuning BERT-based encoders; soft-prompt tuning; learned classification and regression heads for routing
LLM Inference	vLLM, OpenAI API, Ollama; cost and latency measurement
Statistical Methods	Multiple hypothesis testing and FDR control (Benjamini–Hochberg); conformal p-values and conformal novelty detection (AdaDetect, one-class classifiers); distributed detection under Byzantine attacks; black-box adversarial attacks; finite-sample guarantees
Scientific Computing	NumPy, SciPy, pandas, scikit-learn, Matplotlib, OpenCV, Jupyter
Systems & Tools	Git, Linux, AWS, MySQL, MongoDB, Neo4j

Last updated: October 2026